

ITA1407 – Ethical Hacking

CO3 - Assessment Tool 3 – BTL (Analyze)

Penetration Testing – Analyze PRIYA DHARSHINI M(192521051)

Experiment: Analyze Nmap Port Scan Results

Aim:

To analyze the ports of an authorized target system using Nmap and identify the meaning of Open, Closed, and Filtered port states.

Objective:

- To identify open ports on the target.
- To identify closed ports.
- To determine whether ports are filtered by a firewall.
- To understand the security implications of exposed services.

Tool Used:

Nmap – Network Mapper

Target:

Authorized lab system / Virtual Machine

Procedure

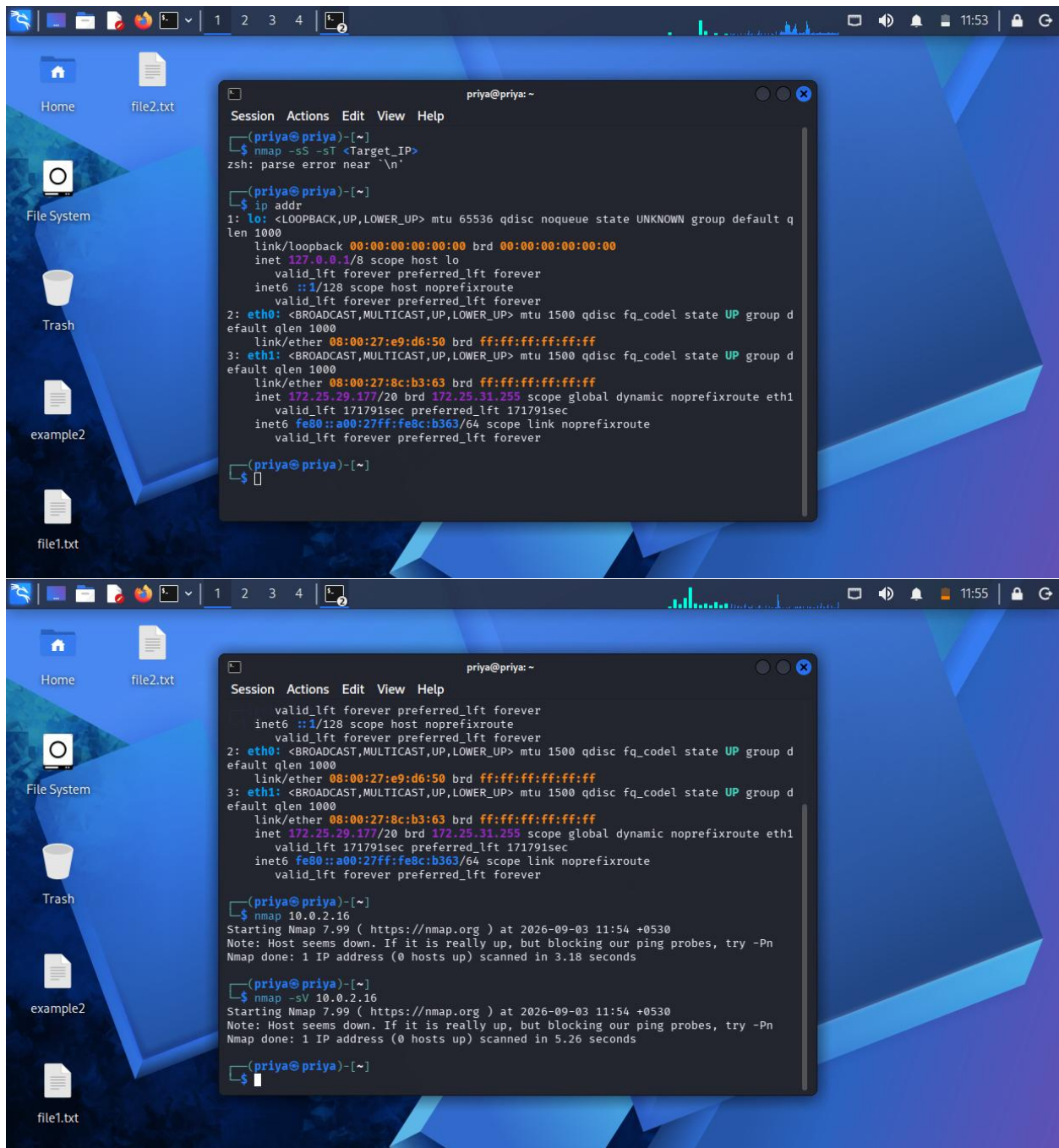
1. Identify the IP address of the authorized target machine.
2. Check whether the target is reachable.
3. Perform a basic Nmap port scan.
4. Perform service/version detection.
5. Analyze the reported port states.
6. Record the security implications of the findings.

Ip addr

nmap 10.0.2.16

nmap -sV 10.0.2.16

nmap -p 22,80,443,8080 10.0.2.16



Observation

Record the output in the following format:

Port	State	Service	Observation
XX	Open	XXXX X	Service is running
XX	Closed	XXXX X	Host is reachable but no service is listening
XX	Filtered	XXXX X	Firewall/filter prevents determining the state

Analysis

Open:

An open port indicates that an application or service is actively listening on that port. Such services should be checked to determine whether they are necessary and securely configured.

Closed:

A closed port indicates that the target host is reachable, but no application is currently listening on that port.

Filtered:

A filtered port indicates that a firewall, ACL, or other network filtering mechanism is preventing Nmap from determining whether the port is open.

Security Assessment

The open ports identified during the scan represent potential attack surfaces. Unnecessary services should be disabled, while required services should be properly configured and updated. Filtered ports indicate that network filtering controls may be in place.

Result

The Nmap scan successfully identified the port states of the authorized target. The results were analyzed to distinguish between Open, Closed, and Filtered ports and to identify potential network attack surfaces.